

Real-World Order of OperationsQuestion Bank 4.07 **ANSWER KEY**

Name

Date

Period

*Show your work. Circle your final answer.***For questions 1 through 4, substitute and evaluate.**

1. $4x^2$ when $x = -3$

36

$4(-3)^2 = 4(9)$

2. $\frac{1}{2}mv^2$ when $m = 50$ and $v = 6$

900

$25(36)$

3. $\frac{a+b^2}{c}$ when $a = 6$, $b = 5$, $c = 2$

 $\frac{31}{2}$ *The bar groups the whole numerator: $6 + 25 = 31$.*

4. $2(x-y)^2$ when $x = 3$ and $y = 7$

32

$2(-4)^2 = 2(16)$

For questions 5 and 6, write and evaluate an expression. Give the unit.

5. A square room of side 11 feet is floored at 3.20 dollars a square foot, with a flat 75 dollars for delivery. Find the total cost.

$11^2(3.20) + 75 = 462.20$ dollars

The rate is per square foot, so it multiplies the area; the delivery is flat, so it does not.

6. A rectangular deck 14 feet by 9 feet is stained at 1.15 dollars a square foot, with a flat 48 dollars for equipment hire. Find the total cost.

$14(9)(1.15) + 48 = 192.90$ dollars

The area is 126 square feet.

Two formulas.

7. Body mass index is $B = \frac{703w}{h^2}$, where w is a weight in pounds and h is a height in inches. Find B to the nearest tenth.

a. $w = 150$, $h = 65$

25.0

$$\frac{105,450}{4,225} = 24.958\dots$$

b. $w = 180$, $h = 70$

25.8

$$\frac{126,540}{4,900} = 25.824\dots$$

c. $w = 200$, $h = 72$

27.1

$$\frac{140,600}{5,184} = 27.122\dots$$

8. Kinetic energy is $K = \frac{1}{2}mv^2$, for mass m in kilograms and speed v in meters per second. Find K in joules.

a. $m = 1,200$, $v = 15$

135,000 joules

b. $m = 850$, $v = 12$

61,200 joules

c. $m = 70$, $v = 8$

2,240 joules

Working backwards.

9. Find the missing value in each situation.

a. $\frac{1}{2}mv^2 = 900$ and $v = 6$. Find m .

$m = 50$

$$\frac{1}{2}m(36) = 900 \text{ , so } 18m = 900 \text{ .}$$

b. $x^2 + 5 = 41$, where x is positive. Find x .

$x = 6$

- c. A square lawn of side n feet is turfed at 2.50 dollars a square foot with a flat 40 dollars delivery, and the total is 2,290 dollars. Find n .

$n = 30$

$$n^2(2.50) = 2,250 \text{ , so } n^2 = 900 \text{ .}$$

Reading the situation.

10. Two problems use the numbers 20, 3 and 5. **A:** a gym charges 20 dollars to join plus 3 dollars for each of 5 classes. **B:** a gym charges 20 dollars for each of 3 months, plus 5 dollars to join. Find both totals, and explain why the words 'plus' and 'each' do not decide the operation.

A is $20 + 3(5) = 35$ dollars. B is $20(3) + 5 = 65$ dollars. Both problems contain the same numbers and the same two words, so the words cannot be what decides; what decides is which amount is repeated.

This is the state guide's named misconception, tested directly: circling the numbers and boxing the key words produces identical markings on two problems with different answers.

11. A square yard of side 15 feet is turfed at 4.80 dollars a square foot, with a flat 120 dollars for delivery. Which expression gives the total cost?

- ☐ A. $15(4.80) + 120$
☒ B. $15^2(4.80) + 120$
☐ C. $15^2(4.80 + 120)$
☐ D. $15^2 + 4.80 + 120$

B

12. Select ALL of the expressions whose value is 45 when $x = 3$.

- ☐ A. $(5x)^2$
☒ B. $\frac{135}{x}$
☒ C. $5x^2$
☒ D. $3x^2 + 18$
☐ E. $5x \cdot 2$
☒ F. $x^2 + 36$

B, C, D and F